

Project Description

Project Title:

Deep learning for medical image diagnosis (AI Radiology Assistant)

Abstract / Summary:

An end-to-end medical diagnostic platform powered by Deep Learning, designed to assist healthcare professionals in identifying and classifying diseases across three distinct imaging modalities: Chest X-Rays, Lung CT Scans, and Brain MRIs. The system features a responsive web interface built using Gradio and integrates a context-aware Large Language Model (LLM) chatbot via the Groq API to deliver personalized medical insights based on the scan results.

Models & Technical Architectures

The framework leverages the **EfficientNet** family of pre-trained convolutional neural networks via Transfer Learning. Each modality utilizes a specific variant optimized for its input spatial resolution and features:

1. Brain MRI Diagnostic Model:

- **Architecture:** EfficientNetB0.
- **Input Resolution:** 224 × 224 pixels.
- **Task:** Multi-Class Classification (4-Class Softmax).
- **Target Classes:** Glioma, Meningioma, Pituitary Tumor, and No Tumor (Normal).

2. Chest X-Ray Diagnostic Model:

- **Architecture:** EfficientNetB3.
- **Input Resolution:** 384 × 384 pixels.
- **Task:** Binary Classification (Sigmoid Output).
- **Target Classes:** Pneumonia vs. Normal.

3. Chest CT Scan Diagnostic Model:

- **Architecture:** EfficientNetB4.
- **Input Resolution:** 380 × 380 pixels.
- **Task:** Binary Classification (Sigmoid Output).
- **Target Classes:** COVID-19 vs. Normal.

Data Preprocessing & Pipeline

- **Image Resizing & Normalization:** Standardizes arbitrary image uploads into exact spatial dimensions and converts them to RGB format.
- **EfficientNet Preprocessing:** Implements native `eff_preprocess` to ensure input pixel distributions align precisely with the distributions the backbone networks were pre-trained on.
- **Logits Handling & Softmax Safety Net:** Includes a specialized calibration function to handle raw logit exports from uncompiled Keras models, eliminating extreme confidence artifacts and ensuring statistically valid probability distributions (0–100%).

Core System Features

- **Interactive Web UI:** Built a lightweight, mobile-responsive frontend utilizing Gradio tabs, allowing seamless multi-page workflow navigation.
- **Dynamic Medical Diagnostics Cards:** Generates color-coded HTML results that immediately signal severity levels (High/Medium/None) alongside specific, actionable clinical descriptions.
- **Custom SVG Data Visualization:** Programmed a zero-dependency SVG donut chart dynamically driven by model confidence percentages to display clear target class distributions at a glance.
- **Context-Aware Medical AI Assistant:** Integrated **Llama 3.3 (70B) via Groq API**. The chatbot is bound by a strict medical system prompt and dynamically updates its state to evaluate the patient's metadata and the latest AI scan diagnosis, giving context-rich explanations.
- **Session History Tracker:** Implements an in-memory session logger that populates a dynamic DataFrame, allowing users to keep track of previous diagnostics in real-time.

Tech Stack:

- **Languages & Frameworks:** Python, TensorFlow, Keras, NumPy.
- **UI & Graphics:** Gradio, HTML5, Custom CSS, Inline SVG.
- **LLM Integration:** Groq API (Llama 3.3 70B).
- **Image Handling:** Pillow (PIL).